ELEC3104: Digital Signal Processing

Topic 6: Multirate Systems

Model Answers

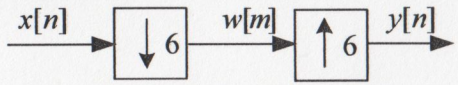
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

Topic 6: Tutorial Questions (SOLO -Level 1)

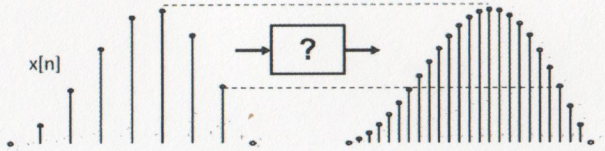
SOLO – Level 1

1. For the diagram shown below, tick the correct answer(s).



a) w[m] = x[6n] ← Down sampling
b) w[m] = x[n/6]
c) y[n] = x[n/6]
d) y[n] = w[m/6] ← upsampling
e) None of the above

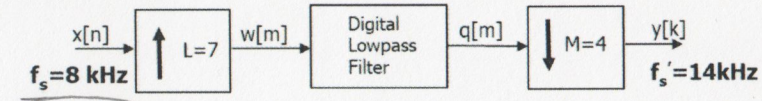
2. What operation is performed on the input sequence x[n] below? (Tick the correct answer)



a) Decimation by 2
b) Decimation by 3
c) Interpolation by 2
d) Interpolation by 3

SOLO – Level 1

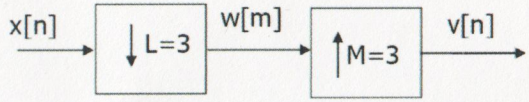
3. Figure below shows sampling rate conversion by non-integer factors. The minimum cut-off frequency of the digital lowpass filter is: (Tick the correct answer)



Bandwidth: 0 to 2 kHz

a) 4 kHz
b) 7 kHz
c) 2 kHz
d) None of the above.

4. For the diagram shown below, tick the correct answer:



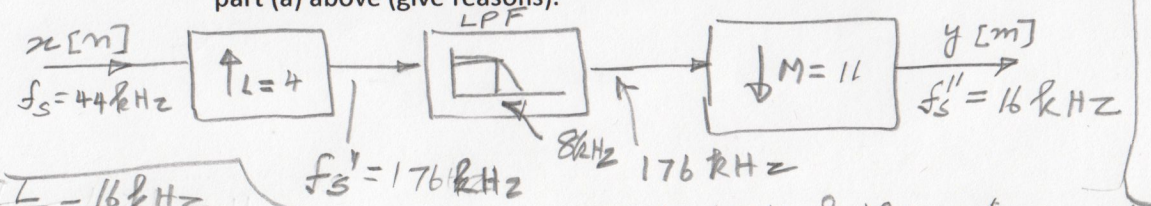
a) w[m] = x[n]
b) v[n]= w[m]
c) v[n]=x[n/3]
d) None of the above

Topic 6: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

1. A signal has been sampled at 44 kHz. For a particular application, this signal is required to be used at a reduced sampling rate of 16 kHz.
   (a) Draw a block diagram to show how you would achieve this.
   (b) Indicate the type of digital filter which you would choose for part (a) above (give reasons).

(a)

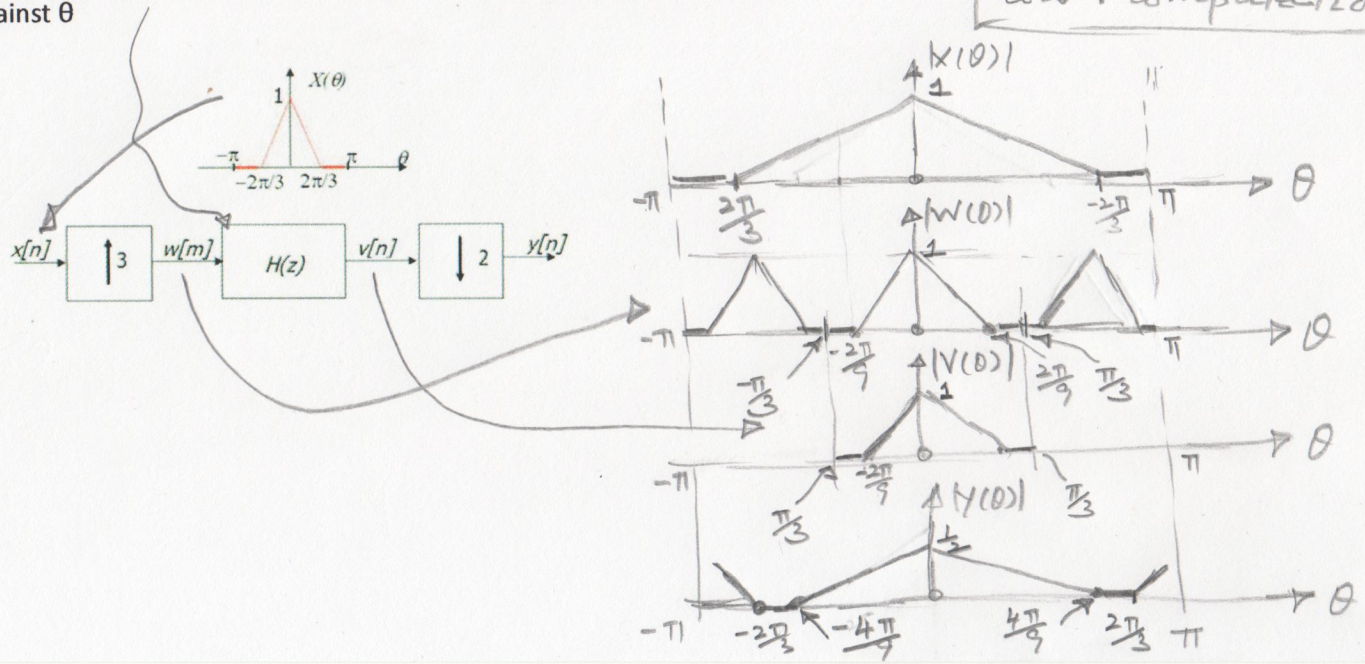


$\frac{L}{M} = \frac{16\,kHz}{44\,kHz} = \frac{4\times4}{4\times11} = \frac{4}{11}$ ⇒ up sample by L = 4 & then down sample by 11

→ The cut-off freq of LPF ≈ $\frac{f_s''}{2}$ = 8 kHz.

(b) Using an IIR filter in this case has an obvious shortcoming. We cannot take advantage of the fact that we have to compute only every M^{th} output, since previous outputs are required to compute M^{th} output. Thus no saving realised. Using an FIR filter implies we can do our computations at f_s'/M since previous outputs are not required to compute the Mth output. This will lead to a significantly lower computation rate

2. A signal x[n] has a spectrum X(θ) as shown below. The signal is applied to the system shown below. The ideal lowpass filter H(z) has a gain factor of 1 in the passband and a cut-off frequency θ=π/3. Sketch |X(θ)|, |W(θ)|, |V(θ)|, |Y(θ)| against θ

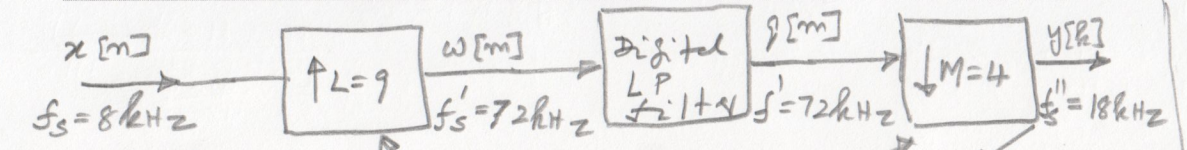


Topic 6: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

3. A signal (with a bandwidth of 0 to 3kHz) has been sampled at 8 kHz and for a particular application this signal is required to be used at an increased sampling rate of 18 kHz:
   (a) Draw a block diagram to show how you would efficiently implement a system to achieve this , indicating the type of digital filter you would use (give reasons);
   (b) Calculate the minimum cut-off frequency of the digital lowpass filter in terms of rad/sec.

(a)



$f_s = 8kHz$

$BW = 0 \text{ to } 3kHz$

$f_s' = 72kHz$

$f_s' = 72kHz$

$f_s'' = 18kHz$

$\frac{L}{M} = \frac{18}{8} = \frac{2\times 9}{2\times 4} = \frac{9}{4}$

$f_s'' = \frac{L}{M} f_s$

$= \frac{9}{4} \times 8kHz$

$= 18kHz$

(b) * The cut off frequency (min) of the lowpass filter must be 3kHz (because the input signal is band limited to 3kHz)

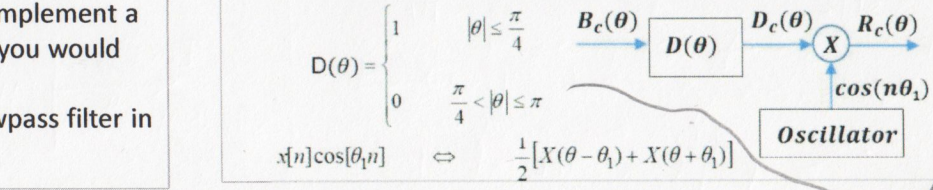
Type of digital filter is FIR

See Question 1(b) answer Page 3

SOLO – Level 2

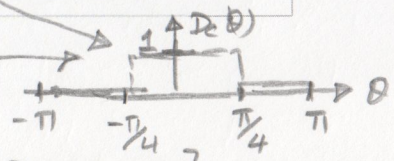
4. A section of a digital modulator is shown below. The input $B_C(\theta) = 1$ for $-\pi \le \theta \le \pi$, By writing appropriate equations for $R_c(\theta)$ and sketch $|R_c(\theta)|$ vs θ, and where $\theta_1 = 3\pi/4$.

$$D(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases}$$



$x[n]\cos[\theta_1 n] \Leftrightarrow \frac{1}{2}[X(\theta-\theta_1) + X(\theta+\theta_1)]$

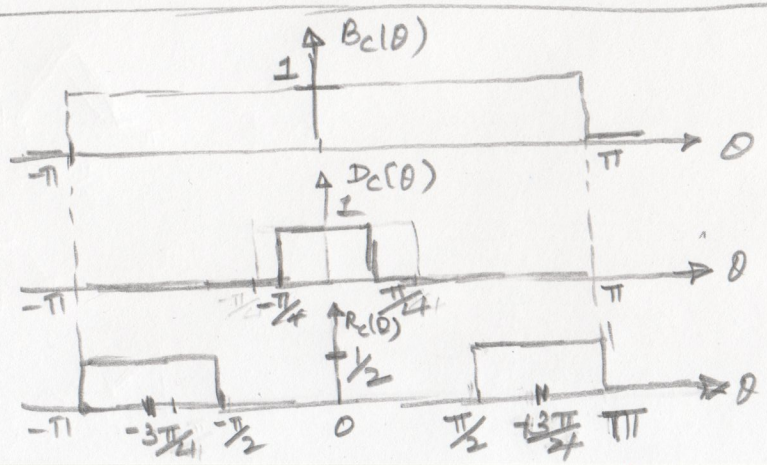
$B_c(\theta) = 1 \; ; \; D_c(\theta) = 1$



$R_c(\theta) = \frac{1}{2}[D_c(\theta-\theta_1) + D_c(\theta+\theta_1)]$

$= \frac{1}{2}[D_c(\theta-\frac{3\pi}{4}) + D_c(\theta+\frac{3\pi}{4})]$

$$R_c(\theta) = \begin{cases} \frac{1}{2} & -\pi/4 \le (\theta - \frac{3\pi}{4}) \le \pi/4 \Rightarrow \pi/2 \le \theta \le \pi \\ \frac{1}{2} & -\pi/4 \le (\theta + \frac{3\pi}{4}) \le \pi/4 \Rightarrow -\pi \le \theta \le -\pi/2 \\ 0 & \text{elsewhere} \end{cases}$$



Topic 6: Tutorial questions (SOLO - Level 3)

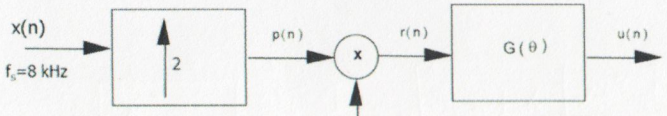
SOLO – Level 3

1. A section of a digital modulator is shown below. If the input signal x[n] is

$$x[n] = \frac{1}{2}\delta[n+1] + \delta[n] + \frac{1}{2}\delta[n-1]$$

(a) Show $P(\theta) = 1+\cos(2\theta)$ for $-\pi \le \theta \le \pi$,

(b) By writing appropriate equations for $R(\theta)$ and $U(\theta)$, sketch $|R(\theta)|$ and $|U(\theta)|$ vs θ, and where $\theta_1 = \pi/4$.



$$G(\theta) = \begin{cases} 1/2 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases}$$

(a) $X(z) = \frac{1}{2}z^{1} + 1 + \frac{1}{2}z^{-1} \quad \therefore X(\theta) = \frac{1}{2}e^{j\theta} + 1 + \frac{1}{2}e^{-j\theta}$

$\therefore X(\theta) = 1+\cos\theta \quad -\pi \le \theta \le \pi$

$\therefore P(\theta) = X(2\theta) = 1+\cos 2\theta$

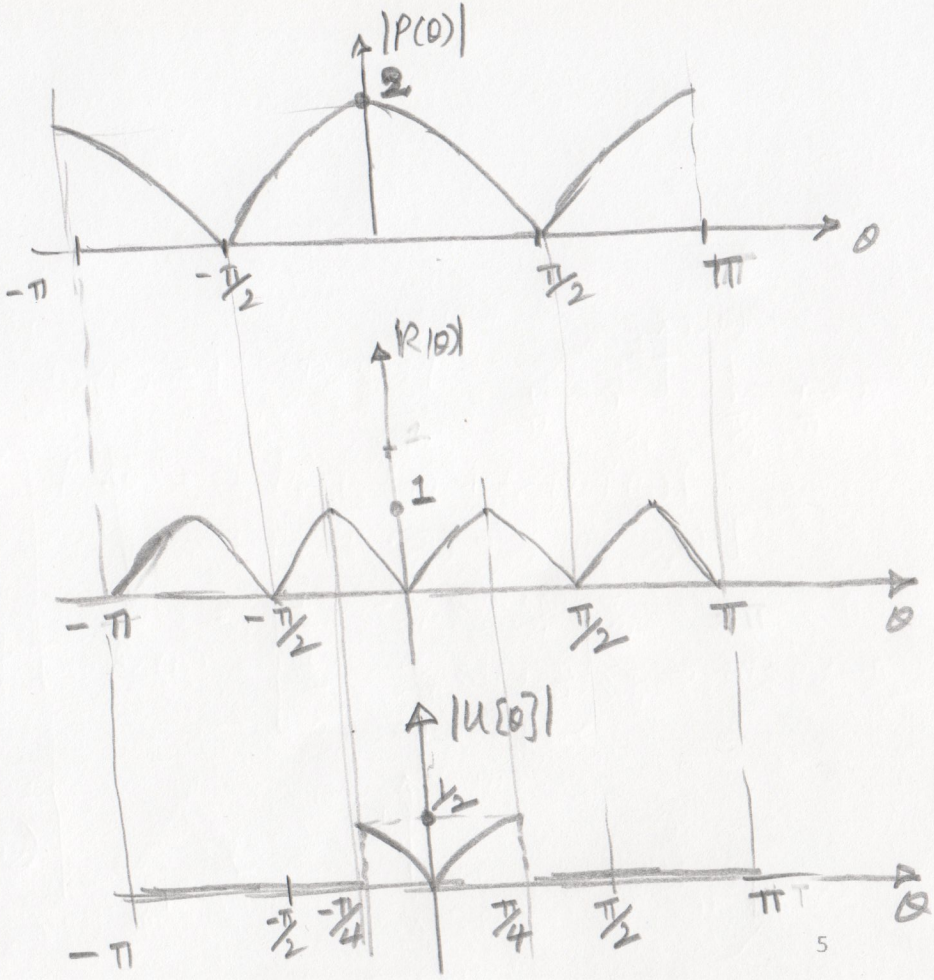
(b) $R(\theta) = \frac{1}{2j}\cdot[P(\theta-\theta_1) - P(\theta+\theta_1)]$

$= \frac{1}{2j}\cdot[1+\cos 2(\theta-\frac{\pi}{4}) - \cos 2(\theta+\frac{\pi}{4}) - 1]$

$= \frac{1}{2j}\cdot[\sin 2\theta + \sin 2\theta] = \frac{1}{j}\sin 2\theta$

$U[\theta] = R[\theta]\cdot G[\theta] = \begin{cases} \frac{1}{2j}\cdot\sin 2\theta & -\frac{\pi}{4} \le \theta \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} \le |\theta| \le \pi \end{cases}$

Given



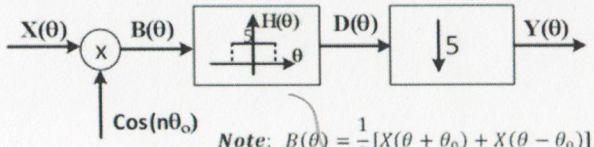
Topic 6: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

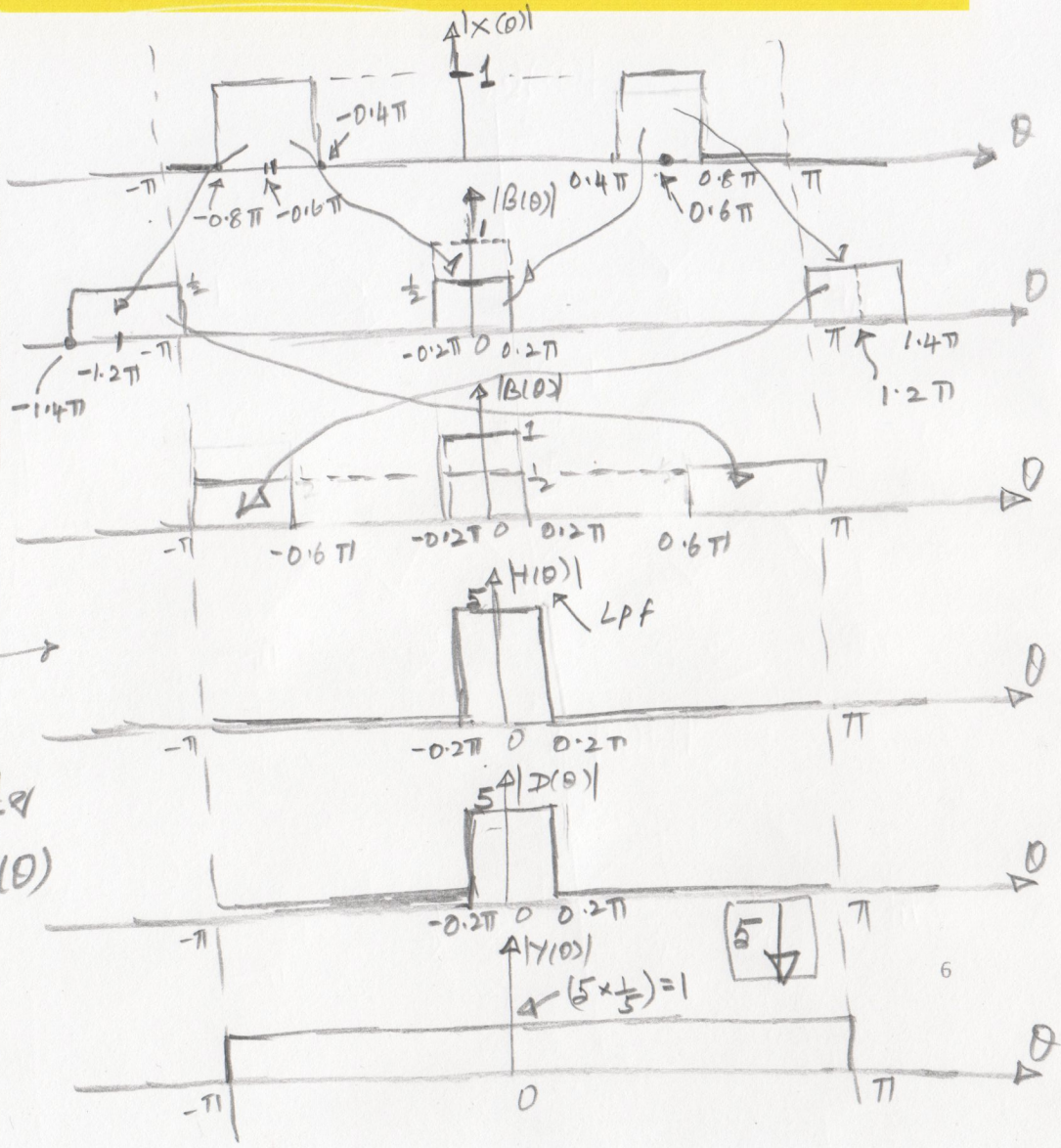
2. The input $X(\theta)$ for the digital modulator shown below is given by:

$$X(\theta) = \begin{cases} 1 & 0.4\pi \le |\theta| \le 0.8\pi \\ 0 & elsewhere \end{cases}$$

The modulator signal frequency is $\theta_o = 0.6\pi$. $H(\theta)$ is an ideal lowpass filter with a cut-off frequency of 0.2π and a gain of 5. Sketch, one above another, $|X(\theta)|$, $|B(\theta)|$, $|D(\theta)|$ and $|Y(\theta)|$ against θ.



Note: $B(\theta) = \frac{1}{2}[X(\theta + \theta_0) + X(\theta - \theta_0)]$



Lowpass filter

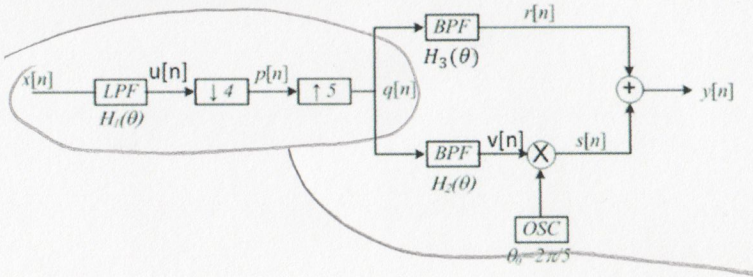
After lowpass filter

$D(\theta) = B(\theta) \cdot H(\theta)$

Topic 6: Tutorial questions (SOLO – Level 4)

SOLO – Level 4

3. A multi-rate system is given below:

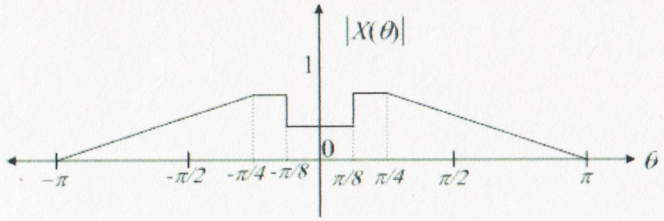


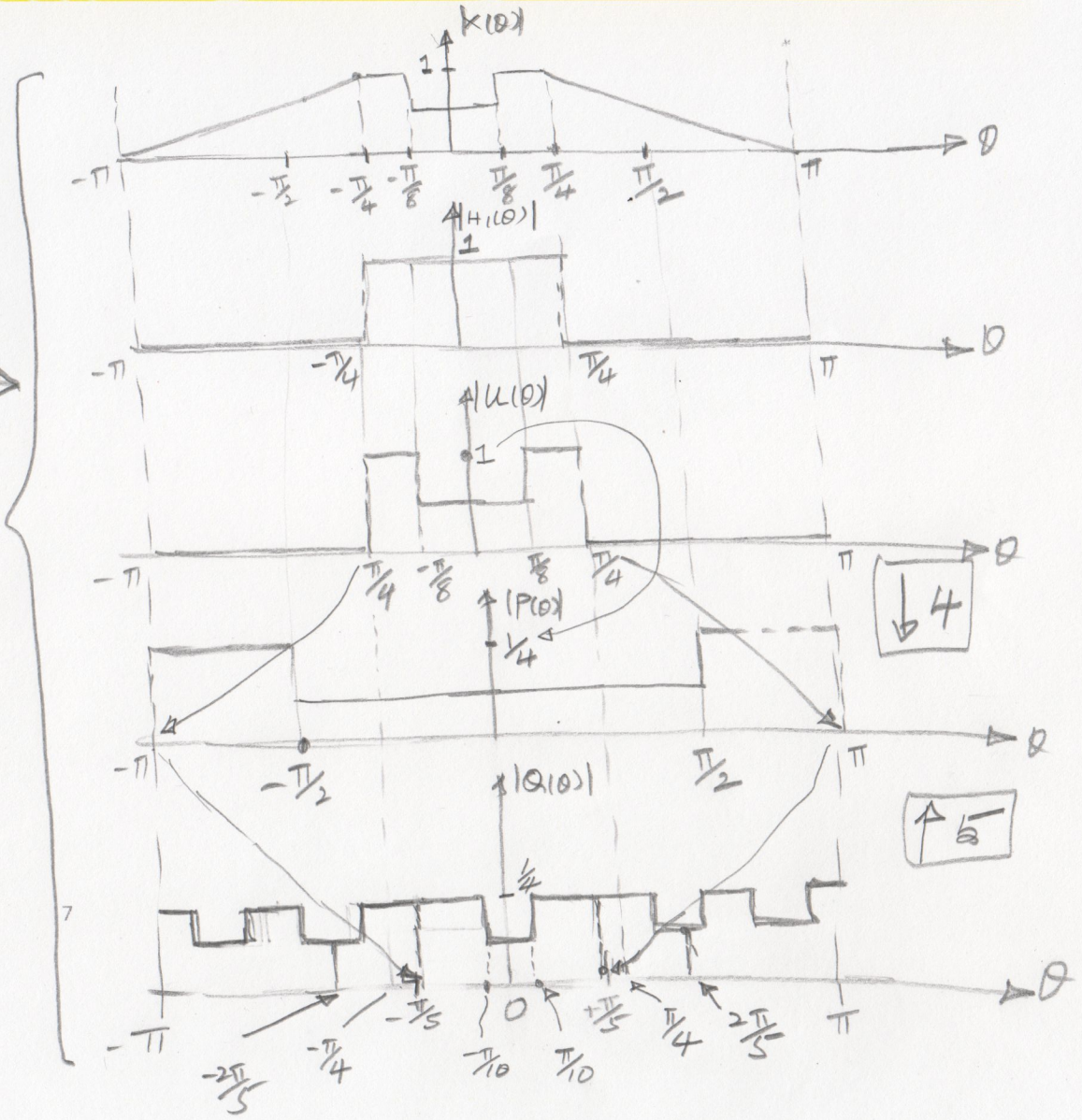
where

$$H_1(\theta) = \begin{cases} 1 & |\theta| \le \pi/4 \\ 0 & otherwise \end{cases} \qquad H_2(\theta) = \begin{cases} 1 & \pi/5 \le |\theta| \le 2\pi/5 \\ 0 & otherwise \end{cases}$$

$$H_3(\theta) = \begin{cases} 1 & 2\pi/5 \le |\theta| \le 3\pi/5 \\ 0 & otherwise \end{cases}$$

If the frequency spectrum of $x[n]$, $|X(\theta)|$ is given as follows, sketch $|P(\theta)|$, $|Q(\theta)|$, $|R(\theta)|$, $|S(\theta)|$ and $|Y(\theta)|$.

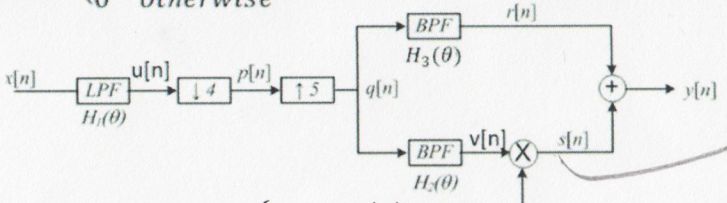




SOLO – Level 4

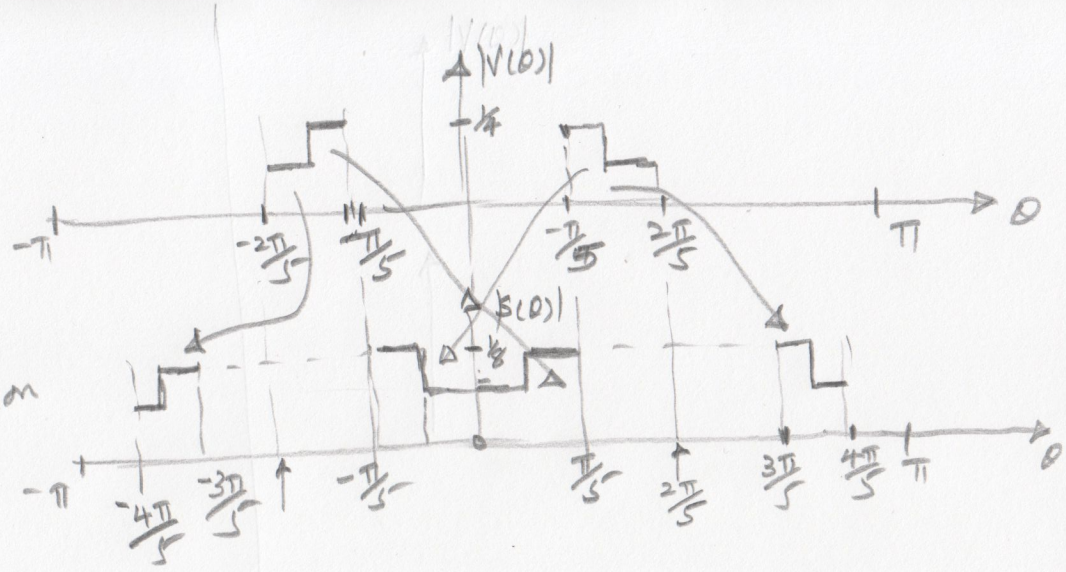
$$H_1(\theta)=\begin{cases}1 & |\theta|\le \pi/4\\ 0 & otherwise\end{cases}$$

$$H_3(\theta)=\begin{cases}1 & 2\pi/5\le|\theta|\le 3\pi/5\\ 0 & otherwise\end{cases}$$



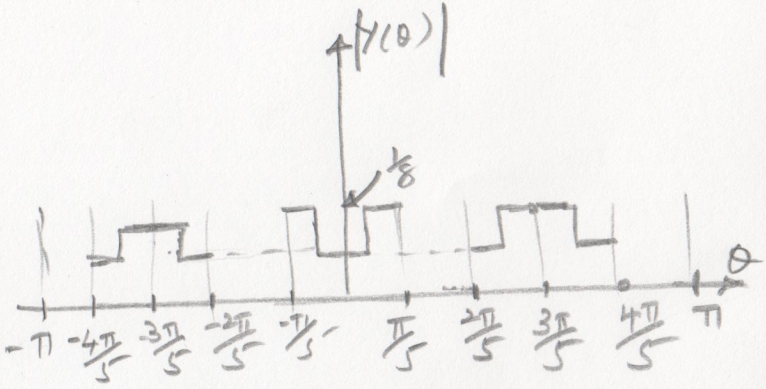
$$H_2(\theta)=\begin{cases}1 & \pi/5\le|\theta|\le 2\pi/5\\ 0 & otherwise\end{cases}$$

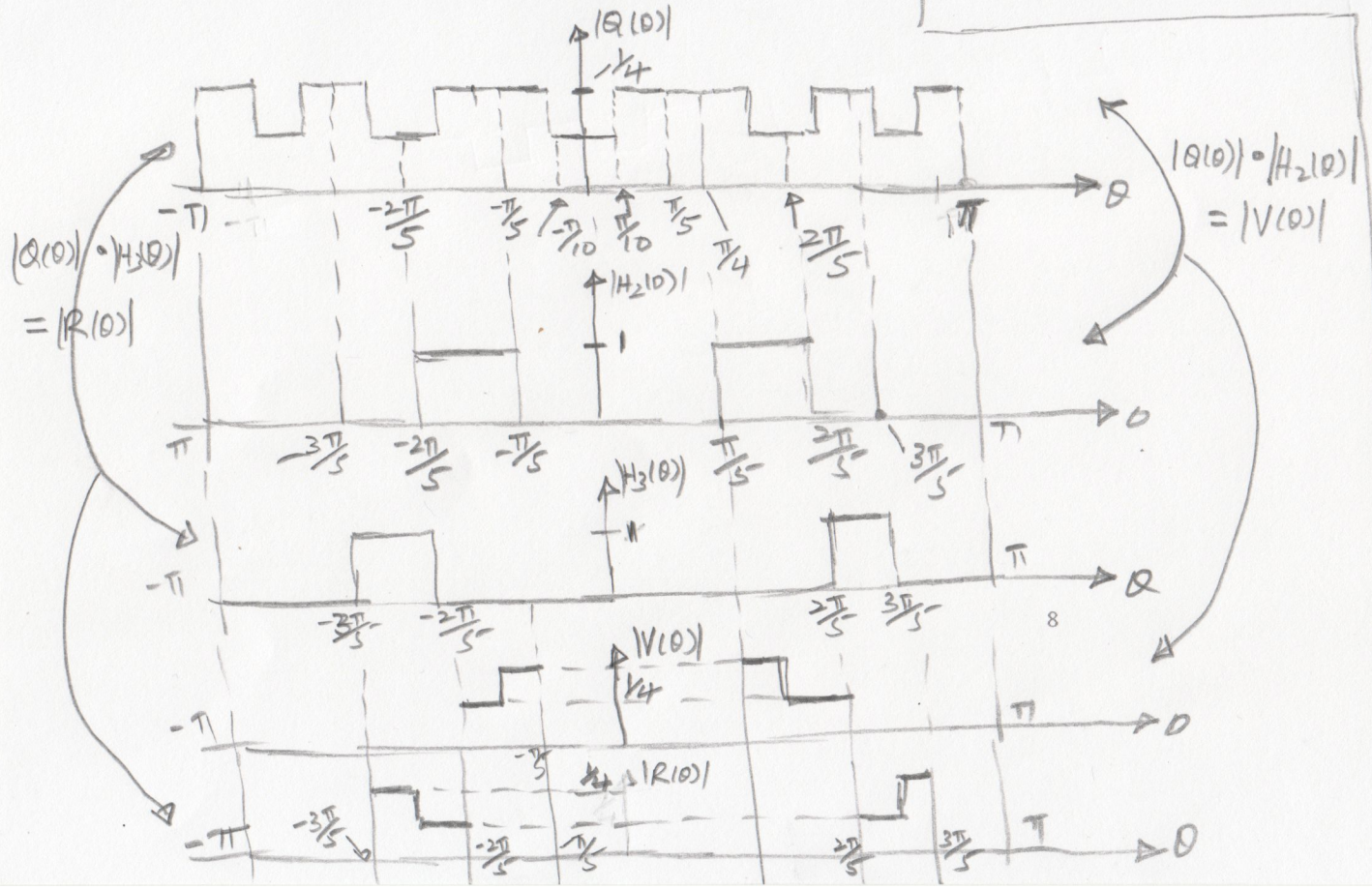
after Modulation



If we now add R(θ) and S(θ), we get

$V(\theta) = S(\theta) + R(\theta)$





Topic 6: Tutorial questions (SOLO - Level 4)

SOLO – Level 4

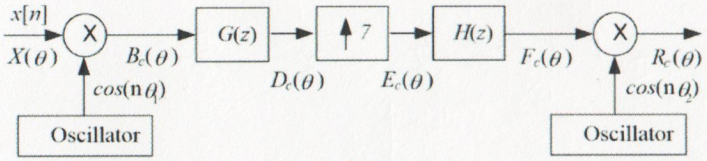
2. The diagram below shows a section of a digital modulator. The sampling period T of the input signal x[n] is 125 μs . The first oscillator generates a carrier of frequency θ_1. The second oscillator generates a carrier of frequency θ_2.

$$\theta_1 = \omega_1 \frac{T}{2} \quad where \omega_1 = 2\pi \times 2 \times 10^3 rad/\sec; \qquad \theta_2 = (\omega_1 + \omega_c)\frac{T}{2}$$

$$where, \omega_c = 2\pi \times 4 \times 10^3 rad/\sec$$

The low-pass filters G(z) and H(z) have the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases} \qquad H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{28} \\ 0 & \frac{\pi}{28} < |\theta| \le \pi \end{cases}$$



(a) If $X(\theta) = 1 + \cos(2\theta)$, by writing appropriate equations, show that

$$D_c(\theta) = \begin{cases} 1 & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere \end{cases}$$

(b) Sketch, one above another, $|E_c(\theta)|$, $|F_c(\theta)|$ and $|R_c(\theta)|$ against θ in the fundamental frequency region $-\pi \le \theta \le \pi$.

$\theta_1 = \omega_1 T/2 = \omega_1 \times \frac{1}{2f_s} = 2\pi \times 2\times10^3 \times \frac{1}{2\times8\times10^3} = \pi/4$

$\theta_2 = (\omega_1+\omega_c)T/2 = \frac{\omega_1+\omega_c}{2f_s} = \frac{2\pi\times2\times10^3 + 2\pi\times4\times10^3}{2\times8\times10^3} = 3\pi/4$

given $\rightarrow X(\theta) = 1+\cos 2\theta$

$B_c(\theta) = \frac{1}{2}[X(\theta-\theta_1) + X(\theta+\theta_1)]$

$= \frac{1}{2}[X(\theta-\pi/4) + X(\theta+\pi/4)]$

$= \frac{1}{2}[\cos 2(\theta-\pi/4) + \cos(2(\theta+\pi/4))]$

$= \frac{1}{2}[1+\sin 2\theta + 1 - \sin 2\theta] = 1$

$D_c(\theta) = B_c(\theta) \cdot G(\theta) = \begin{cases} 1 & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere. \end{cases}$

$B_c(\theta) = 1$ ↑ given

without taking equation you can sketch the spectrum.

$F_c[\theta] = D_c(7\theta) \circ H(\theta) = \begin{cases} \begin{cases} 1 & -\pi/4 \le 7\theta \le \pi/4 \\ 0 & \pi/4 \le |7\theta| \le \pi \end{cases} \\ 0 & \frac{\pi}{7} \le |\theta| \le \pi \end{cases}$

$= \begin{cases} 1 & -\frac{\pi}{28} \le \theta \le \frac{\pi}{28} \\ 0 & \frac{\pi}{28} \le |\theta| \le \pi \end{cases}$

$R_c(\theta) = \frac{1}{2}[F_c(\theta-\theta_2) + F_c(\theta+\theta_2)]$

$= \frac{1}{2}[F_c(\theta-3\pi/4) + F_c(\theta+3\pi/4)]$

$= \begin{cases} 1/2 & -\pi/4 \le \theta - 3\pi/4 \le \pi/28 \\ 1/2 & -\pi/28 \le \theta + 3\pi/4 \le \pi/28 \\ 0 & elsewhere \end{cases} = \begin{cases} 1/2 & 10\pi/14 \le \theta \le 11\pi/14 \\ 1/2 & -11\pi/14 \le \theta \le -10\pi/14 \\ 0 & elsewhere \end{cases}$

Topic 6: Tutorial questions (SOLO - Level 4)

SOLO – Level 4

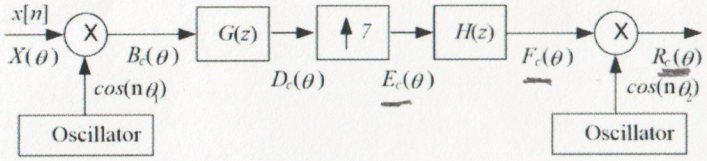
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$$\theta_1 = \omega_1 \frac{T}{2} \quad where \omega_1 = 2\pi \times 2 \times 10^3 rad/\sec; \qquad \theta_2 = (\omega_1 + \omega_c)\frac{T}{2}$$

$$where, \omega_c = 2\pi \times 4 \times 10^3 rad/\sec$$

The low-pass filters G(z) and H(z) have the following characteristics:

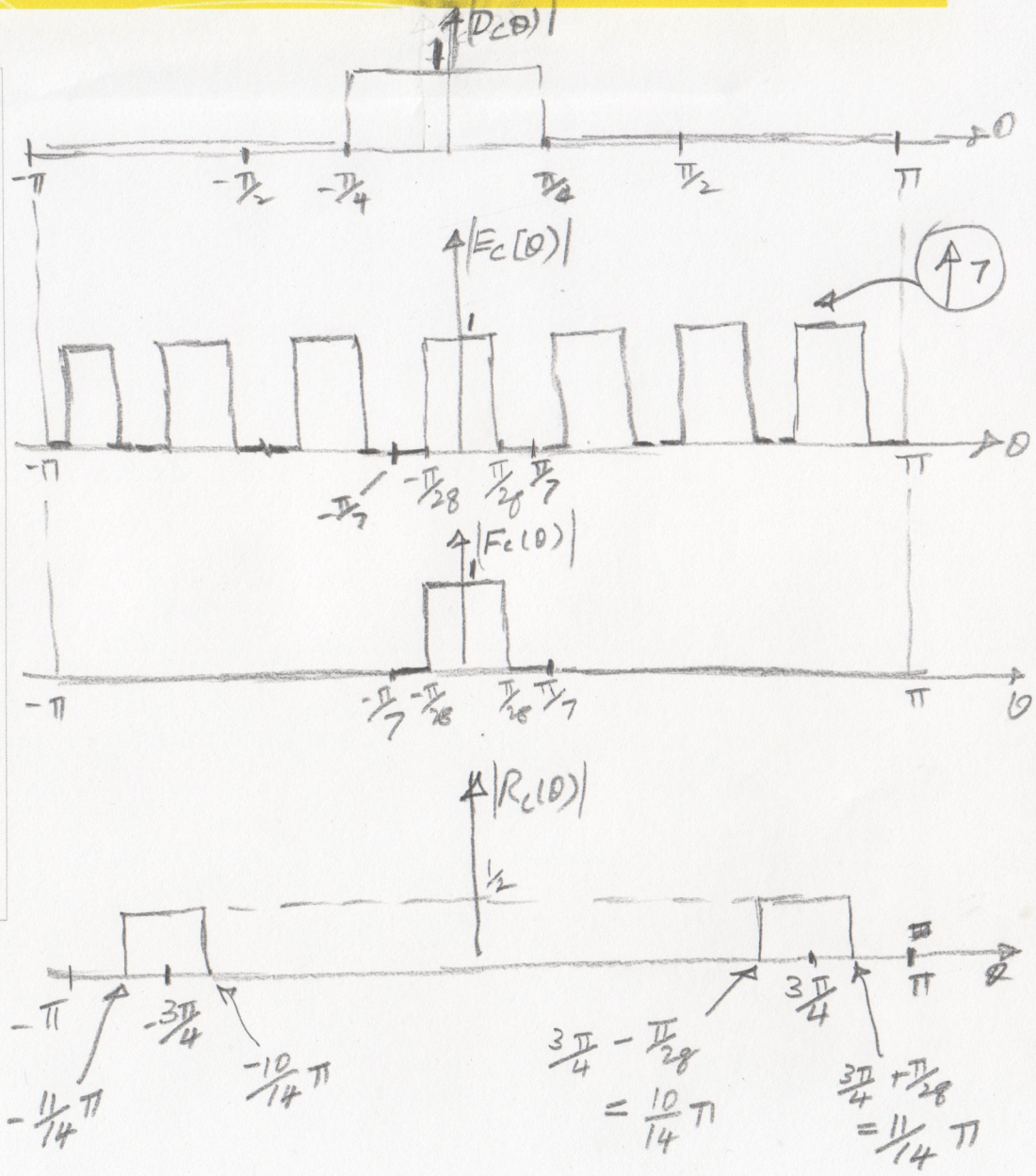
$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases} \qquad H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{28} \\ 0 & \frac{\pi}{28} < |\theta| \le \pi \end{cases}$$



(a) If $X(\theta) = 1 + \cos(2\theta)$, by writing appropriate equations, show that

$$D_c(\theta) = \begin{cases} 1 & -\pi/4 \le \theta \le \pi/4 \\ 0 & elsewhere \end{cases} \quad \checkmark$$

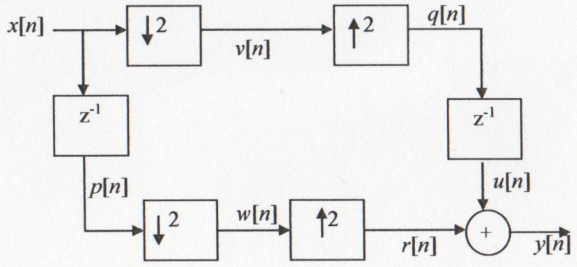
(b) Sketch, one above another, $|E_c(\theta)|$, $|F_c(\theta)|$ and $|R_{c(\theta)}|$ against θ in the fundamental frequency region $-\pi \le \theta \le \pi$.



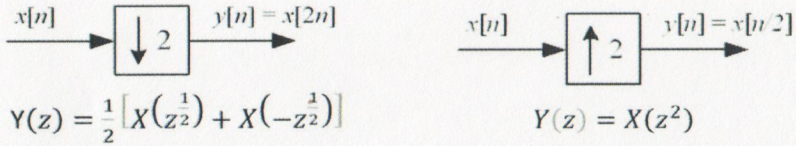
Topic 6: Tutorial questions (SOLO - Level 5)

SOLO – Level 5

1. Express the output y[n] of the figure below as a function of the input x[n]. By simplifying the expression derived, show that y[n] = x[n-1].



Note:



$p[n] = x[n-1] \Rightarrow P(z) = X(z)\,z^{-1}$

$P(z^{\frac{1}{2}}) = X(z^{\frac{1}{2}}) \cdot z^{-\frac{1}{2}} \Rightarrow P(-z^{\frac{1}{2}}) = X(-z^{\frac{1}{2}})(-1)\,z^{-\frac{1}{2}}$

$W[n] = p[2n] \Rightarrow W(z) = \frac{1}{2}\left[P(z^{\frac{1}{2}}) + P(-z^{\frac{1}{2}})\right]$

$\therefore W(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) \cdot z^{-\frac{1}{2}} - X(-z^{\frac{1}{2}})\,z^{-\frac{1}{2}}\right]$ — (1)

$V[n] = x[2n] \Rightarrow V(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}})\right]$ — (2)

$q[n] = V[\frac{n}{2}] \Rightarrow Q(z) = V(z^2)$

$U(z) = Q(z) \cdot z^{-1}$

similarly $R(z) = W(z^2)$

We have:

$Q(z) = V(z^2) = \frac{1}{2}\left[X(z) + X(-z)\right]$ (from 2)

$R(z) = W(z^2) = \frac{z^{-1}}{2} X(z) - \frac{z^{-1}}{2} X(-z)$ (from 1)

$U(z) = z^{-1} Q(z) = \frac{1}{2} z^{-1}\left[X(z) + X(-z)\right]$

$\therefore Y(z) = U(z) + R(z)$

$= \frac{z^{-1}}{2} X(z) - \frac{z^{-1}}{2} X(-z) + \frac{1}{2} z^{-1}\left[X(z) + X(-z)\right]$

$\therefore Y(z) = X(z)\,z^{-1}$

$y[n] = x[n-1]$

↑ The original signal x[n] is reconstructed. The above structure is a perfect reconstruction system.